Senior Engineer, Scientific Software and Developer Platform, Modeling and Simulation

Kairos Power · Remote or Alameda, CA

Job ID (this archive)	ENERGY-093
Employer	Kairos Power
Location(s)	Remote or Alameda, CA
Career stage (archive tag)	Experienced (Senior)
Location group	Other U.S.
Majors (archive tags)	Computer Science, Data Science
Energy area	Nuclear / scientific computing
Pay listed	$140,611 – $165,425 per year
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/kairospower/jobs/6163824004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-064

Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Company Overview

Kairos Power is a new nuclear energy technology and engineering company whose mission is to enable the world's transition to clean energy, with the ultimate goal to dramatically improve people's quality of life while protecting the environment. This goal will be accomplished through the commercialization of the fluoride-salt-cooled, high-temperature reactor (FHR) that can be deployed with robust safety, affordable costs, and flexible operation to accommodate the expansion of variable renewables.

Job Summary

Kairos Power is looking for a Senior Scientific Software and Developer Platform Engineer to work within a small software and infrastructure team that supports the needs of our design, research, and development organizations. This is a hands-on individual contributor role. You will architect and build the scientific software, developer tooling, and delivery systems that nuclear engineers and computational scientists use to design and analyze nuclear power systems.

The work spans Python and C++ development, computational modeling, build and packaging systems, reproducible artifact delivery, CI/CD, containerization, software quality assurance, and high-performance computing workflows. You will set software architecture direction, establish the design and maintainability standards the group works to, and raise engineering practice through design review, code review, and mentoring. The group favors simple, proportionate designs assembled from standard, well-supported technologies, and this role is expected to be a leading voice for that standard. Success means writing and shaping software that engineers other than its original author can operate, extend, and maintain.

Responsibilities

Scientific and engineering software

- Design, implement, test, and maintain scientific and engineering software in Python and C++, including the implementation of computational modeling methods and numerical algorithms.
- Own software architecture for internally developed engineering tools, selecting data structures, interfaces, and algorithms that match the actual problem rather than an anticipated one.
- Profile and optimize performance where a measurable requirement justifies the work, and document the resulting characteristics.

- Collaborate with nuclear and other non-software engineers to define software projects based on their needs for tools that support the primary mission of the company.

Build, packaging, and artifact delivery

- Own CI/CD for internally developed software using GitHub Actions or comparable systems, including automated testing, quality gates, and controlled release.
- Build reproducible build and packaging pipelines for Python packages, Conda environments, and C++ binaries and libraries, including compiler and dependency management.
- Own artifact publication and versioning, so that consumers can determine exactly what they are running and reproduce it later.
- Diagnose and resolve compilation, dependency, environment, and runtime problems across the scientific software stack.
- Establish and maintain container build and runtime patterns for Docker and Apptainer across development and high-performance computing use, and own base image currency and upgrade paths so that containers stay patched without breaking the workflows built on them.

Software quality assurance and controlled delivery

- Uphold internal software quality assurance requirements, and build the automation that makes compliance a property of the pipeline rather than a manual step.
- Define the artifacts, records, and gates required for controlled software delivery, and make them reproducible from the build system.
- Support software quality audits with evidence generated by tooling rather than assembled by hand.

Scientific computing platform

- Develop and support high-performance computing user workflows, including Slurm scheduling, Open OnDemand interactive access, MPI, and GPU computing.
- Improve the day-to-day experience of the engineers and scientists who use these platforms, and reduce the friction in their modeling and simulation workflows.
- Build self-service engineering workflows and documented paths that let users solve common problems without opening a request.

Engineering standards and mentorship

- Establish design, documentation, and maintainability standards that other teams use to develop and deploy code, and apply them consistently through technical design review and code review.
- Review software produced across the group, including internal applications, so that no system depends on a single author.
- Mentor engineers, transfer knowledge deliberately, and remove single-person dependencies through documentation and shared ownership.
- Evaluate build-versus-buy decisions, and select standard, well-supported libraries and tools over custom frameworks unless a requirement clearly demands otherwise.
- Choose the smallest design that meets the actual requirement, and identify and retire technical debt, premature abstraction, and unnecessary complexity.
- Participate in the team’s engineering help queue for issues that require deep software expertise, and communicate resolutions clearly to the engineers who reported them.

Qualifications

- Bachelor’s degree in computer science, engineering, physics, mathematics, computational science, or a related technical field, or equivalent combination of education and practical experience.
- 5+ years of relevant software engineering experience, including experience developing scientific, engineering, computational, or other technically complex software.
- Strong and demonstrable experience developing scientific or engineering software, with depth in both Python and C++.
- Experience implementing computational modeling methods, numerical methods, or comparable quantitative algorithms.
- Demonstrated software architecture experience, including the ability to decompose a problem, define interfaces, and reason about maintainability, performance, and long-term cost.
- Experience with automated testing, including unit and integration testing with frameworks such as PyTest or GoogleTest.
- Proficiency with Git, Linux, Bash, and SSH.
- Experience mentoring engineers and conducting design and code review, together with a demonstrated ability to accept and apply design feedback.

- A track record of building software that other engineers can operate, extend, and maintain, supported by clear documentation and deliberate knowledge transfer.

Desired Qualifications

- Master's or Ph.D. in computational science, physics, engineering, or a related discipline, particularly with research experience implementing numerical methods is desirable.
- Experience with containerization using Docker, Apptainer, or comparable tools is desirable.
- Experience with Python packaging and environment management, including Conda, and with versioned artifact publication is desirable.
- Experience with build systems such as CMake, and with compiler and dependency management for compiled software is desirable.
- Experience with CI/CD systems such as GitHub Actions, including pipeline design and release automation is desirable.
- Experience with performance profiling and optimization driven by measurement rather than assumption is desirable.
- Experience with software quality assurance practices and controlled software delivery is desirable.
- Experience with high-performance computing environments, including Slurm, Open OnDemand, MPI, and GPU computing is desirable.
- Experience with software quality assurance in a regulated or safety-conscious engineering environment is desirable.
- Experience with documentation tooling such as Doxygen or Sphinx is desirable.
- Experience with relational databases, backend APIs, or cloud platforms such as AWS or Google Cloud is desirable.
- Familiarity with infrastructure as code and the operational side of software delivery is desirable.

Knowledge, Skills & Abilities

- Proficiency with Git, Linux, Bash, and SSH.
- Strong command of software design fundamentals, including object-oriented programming, data structures, and algorithms.
- Ability to systematically diagnose complex software, build, dependency, environment, and runtime issues and identify root causes.
- Ability to design and maintain reproducible software build, test, packaging, and delivery processes.
- Ability to apply software quality assurance principles, including traceability, testing, documentation, and controlled delivery, appropriate to the software being developed.
- Ability to communicate effectively with nuclear engineers, computational scientists, and other non-software specialists, and to translate their requirements into working software.
- Ability to work with a highly collaborative team.
- Ability to solve problems quickly and efficiently.
- Prioritizes and ensures safety of oneself and others.
- Ability to proactively collect, manage and transfer knowledge.
- Ability to create efficiencies when working on problems, challenges, and issues, where each improvement is tied to a measurable engineering need such as safety, reliability, performance, usability, or maintainability.
- Ability to exercise architectural judgment and to favor the simplest design that satisfies the requirement.
- Ability to give direct technical feedback, to receive it, and to change course when a review identifies a better approach.
- Ability to explain technical tradeoffs to engineers and scientists who are not software specialists.
- Ability to take operational ownership of software through its full lifecycle, including testing, documentation, support, and retirement.

Physical Conditions

- Remaining in a stationary position, often standing, or sitting for prolonged periods
- Moving about to accomplish tasks or moving from one worksite to another
- Communicating with others to exchange information

Environmental Conditions

- Remote work
- General office environment
- High-concentration, demanding and fast-paced.

Safety and PPE

- Reading and interpreting hazardous warning signs
- Manipulating, cleaning and disposal of hazardous materials

Travel

- Some travel may be required for workshops and project assistance.

Additional Requirements

- Occasionally requires working weekends.
- Occasionally requires schedule flexibility.
- Occasionally requires extended hours to support launch and critical project timelines.

The salary range for this position is $140,611 to $165,425 for employees working onsite at our Alameda, CA headquarters. The range provided in this job posting represents the typical range or starting rate of candidates hired in California. Factors that maybe used to determine your actual salary may include your education, experience, knowledge, skills, ability and the market data for your work location, and a comparison to other employees already in the role.

#LI-Remote

About our Benefits

We know that we have some of the most talented and dedicated employees, and we believe in rewarding them accordingly. If you work here, full-time employees (excludes interns) expect to have access to the benefits below:

- Competitive compensation packages
- Medical, dental and vision benefits for employees and their dependents
- Paid Vacation
- 401(k) and pre-tax health insurance, dependent care, and commuter benefits (FSA)

Kairos Power is committed to building a diverse workforce that reflects the communities where we do business. Kairos Power is an equal opportunity employer. All qualified applicants will receive consideration for employment without regard to race, color, religion, gender, gender identity or expression, sexual orientation, national origin, genetics, disability, age, or veteran status. We actively welcome people of different experiences, abilities, and perspectives.

Kairos Power participates in E-Verify.

Candidates only, no recruiters or agencies please.

Note: You are applying for a position that is located in a facility that handles information that is subject to export control restrictions by the Department of Energy under 10 CFR Part 810. To work in this facility, you need to be authorized by the Department of Energy to access Part 810-controlled information. Foreign nationals who are citizens of countries that are not on the Department of Energy's general authorization list (link below) are not permitted to work in our facility unless the Department of Energy issues an export control license to the company to permit that individual to have access to Part 810-controlled information. The following questions are intended to determine the licensing requirements that may apply.
https://www.law.cornell.edu/cfr/text/10/appendix-A_to_part_810

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Bachelor's degree in computer science, engineering, physics, mathematics, computational science, or a related technical field, or equivalent combination of education and practical experience. | 5+ years of relevant software engineering experience, including experience developing scientific, engineering, computational, or other technically complex software. | Master's or Ph.D. in computational science, physics, engineering, or a related discipline, particularly with research experience implementing numerical methods is desirable.

Key skills listed in the archive: Python; C++; Linux; Docker; AWS; Git; optimization; genetics; automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6163824004

Career stage: Senior

Major fit: Direct skills pathway: software development, computing and engineering; consult the exact degree and experience requirements. Energy-sector connection: Scientific software and simulation for advanced nuclear energy.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6163824004.

Sources and retrieval details

Employer posting

https://job-boards.greenhouse.io/kairospower/jobs/6163824004

Retrieved source

https://boards-api.greenhouse.io/v1/boards/kairospower/jobs?content=true

Retrieved: 2026-10-26T16:33:09.192778+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.